

Y2 Electromagnetism 2

Non-assessed problem answers 3 (weeks 5–6)

Magnetic field in materials

- 1) In the first case, $B_{\text{out}} = B_{\text{in}} = 0$, $H_{\text{out}} = 0$, $H_{\text{in}} = -M$.
In the second case, $B_{\text{out}} = 0$, $B_{\text{in}} = \mu_0 M$, $H_{\text{out}} = 0$, $H_{\text{in}} = 0$.

2)

$$N = \frac{B}{\mu_0 I} \cdot \left(\frac{L}{\mu} + h \right) = 3016.$$

3) See lecture 1.

4) A proof is available in the full solutions.

5) At point A,

$$B_A = \mu_0 M \frac{\Omega}{4\pi} = \mu_0 M \left(1 - \frac{R^2}{2L^2} \right),$$

where Ω is the solid angle subtended by the side surface of the magnet. At point B,

$$B_B = \frac{\mu_0 M}{2}.$$

At point C, using the boundary conditions,

$$B_C = B_A - \mu_0 M = -\mu_0 M \frac{R^2}{2L^2}.$$

6) For $R \ll L$, $B_{\text{in}} = \mu B_0 \gg B_0$. For $R \gg L$, $B_{\text{in}} = B_0$.

7) Inside the dielectric material,

$$B = \frac{\varepsilon - 1}{\varepsilon} \cdot \frac{\mu_0 \lambda \omega}{2\pi}.$$

Outside the dielectric material (in both inner and outer regions), $B = 0$.

8) The magnetic field lines are concentric circles;

$$B(r, t) = \varepsilon_0 \varepsilon \mu_0 \mu \cdot \frac{\omega r}{2L} \cdot V_0 \cos(\omega t).$$